Frameworks for LLM Scoring

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1 Comparative Scoring

1.1 Motivation

While LLMs seem great at reasoning, they are much worse than traditional ML models at producing accurate numeric scores. In the context of trading, this means that we should not ask LLMs questions of the form "how much will Apple stock go up today". Raw numeric outputs from LLMs are typically biased, poorly calibrated, and context specific. Even if an LLM says that Apple Stock may go up by 0.2 percent in the next trading day, we do not have enough data or conviction to treat this numeric output as an actual forecast (in the way that one might with a model like XGboost or a Deep Neural network).

However, one should not discard these forecasts entirely; if an LLM agent who has read the entirety of today's news believes that they will favor Apple over the next few trading days, I will be inclined to buy Apple.

The question though is how much? And how does buying Apple compare to buying Microsoft, or NVDIA? How do all of the recent world developments affect each of these stocks relative to each other?

1.2 Ranking

Relative comparisons are much easier for LLMs than numeric forecasts. Given a set of 10 stocks, and a large amount of associated context and news, how will they perform relative to each other? LLMs are much better suited for answering questions of this type. It's also a much easier question; the total space of outcomes is just the total number of permutations of the stocks. It's true that in a universe of even just 1000 stocks this number can blow up. But we can simplify the problem significantly by splitting similar stocks into smaller buckets that each get sent to an LLM ranker. This has 2 benefits - the first is that by binning in this way, we can prevent any individual LLM ranker from ranking solely based on common market factors. Second, we reduce the space of things for the LLM to reason over.

Concretely, our current LLM ranking system does the following.

- Load context that is relevant across the entire universe (macro data, geopolitical and financial news): C. This must come from some lookback window t
- Bin universe by the top 10 risk factors (momentum, industry...), leading to b buckets of size n
- For each bucket, load specific news and context relevant to the stocks in that bucket
- Have the LLM output a concrete ranking of the n stocks in its bucket, using only the shared and specific context

1.3 Bradley-Terry

Once we've accumulated a set of rankings, the next step for our strategy is to convert these into an actual set of scores. As noted in our previous paper, *Towards an AI-Native Trading Strategy*, we use these scores as "views" that are to be combined with a market-cap prior in the Black-Litterman algorithm

A well-studied and popular approach here is known as Bradley-Terry. This algorithm is best applied in situations like chess or tennis, where there are m players, each with their own strengths. We only ever observe pairwise matchups between players, which form the m x m adjacency matrix W, where W_{ij} represents the number of times player i beat player j. For the purposes of our analysis here, the diagonal entries (i.e each players' comparison with themselves) are zero

Bradley-Terry then says that the probability that player i beats player j in any particular matchup is the following:

$$P(i \succ j) = \frac{\pi_i}{\pi_i + \pi_j} \tag{1}$$

Where π_i is e^{B_i} and B_i represents the strength of player i. Given only the pairwise comparisons from the matrix W, it can be shown that an iterative algorithm converges to the MLE estimate for the scores B_i, under a few constraints. The iterative update is defined as follows:

$$\pi_i^{(t+1)} = \frac{w_i}{\sum_{j \neq i} \frac{n_{ij}}{\pi_i^{(t)} + \pi_j^{(t)}}} \tag{2}$$

where w_i is the total number of wins by player i and n_{ij} is the total number of games played between player i and j.

Note that there are also few additional assumptions needed to really make the B-T model work.

1. Given a set of all players' scores, if we multiply them by a constant c, the probabilities do not change. Therefore, there exists a whole set of solutions to B-T. In order to get around this, we often introduce the constraint $\Sigma\pi_i = 1$

2. Less obviously, another problem arises in the case in which there is one player, p_w, who beats everyone else. Because the scores are continuous we can then improve the probability $P(p_w > p_i)$, for all i, by simply doubling the score of p_w and then re-normalizing. To get past this issue, we also require the (modest) assumption that for any splitting of the total players m into two groups A amd B, there always exists a player in B who beats a player in A. This condition is equivalent to saying that the DAG formed from the adjacency matrix W is strongly connected (where the directed edge from $i- > j$ indicates that player i has beatend player j at least once

The first constraint is easy to satisfy for our LLM scoring framework; we re-normalize after every update step. The second condition (known as Ford's condition), is a bit harder in guarantee. In our case, this would be having an asset in our universe that ranks first in every subset that it is in; it never loses. In practice, this has a very low probability of occurring. In our implementation of the algorithm, there is also a parameter for the number of groups a stock is sent to be ranked. Increasing this number decreases the chance of an "undefeated" stock exponentially.

1.4 Proof of Convergence

Here we sketch out a rough outline for the convergence of the update algorithm given in (2). The algorithm given is named after the German mathematician Zermelo who first studied the algorithm in detail in the 1920s. It is an instance of a Minorization Maximization algorithm (MM), which is a class of optimization algorithms (of which the more commonly known Expectation Maximization is a special case).

MM algorithms exist to find optima in cases where the objective function is not easy to solve. The general idea is to find a lower bound on the actual objective function (minorization) at our current guess of the optima, maximize that surrogate function to obtain a new guess of the optima, and then repeat. In the case of B-T, we seek to maximize the Maximum Likelihood of our data, and our parameters are the scores assigned to all the players.
For a given player i, the log-likelihood of seeing the data is given by this equation:

$$\Sigma w_i[\log(\pi_i) - \log(\pi_i - \pi_j)] \tag{3}$$

where the summation is over i, j. For a particular player i, we can just sum over the other players j (being careful with the fact that the cross term shows up twice).

The problematic term here is the cross-term, so we seek to find a minorization of it via the tangent line to $ln(x)$. The tangent line at any particular point x is equivalent to $ln(x) + ln'(x)(x' - x)$ where x' is the new point from which we are evaluating from. Therefore, we can eliminate the problematic term in (3) by using the following inequality:

$$-\log(\pi_i - \pi_j) \geq -\log(\pi_i - \pi_j) + \frac{(\pi_i' - \pi_j')}{(\pi_i - \pi_j)} - 1 \tag{4}$$

with equality only at our current point $\pi_i - \pi_j$. Once we've pulled the $\pi_i - \pi_j$ from the *log*, we can find the derivative of the log-likelihood w.r.t to any single player easily and perform a one-step update. it can be then shown that this results in the classic Zermelo algorithm. In practice, a "cyclic" algorithm is performed, where the updated score for a player is immediately used for the update step of the subsequent player.

2 Scoring Frameworks

We now turn our attention to more general scoring frameworks. B-T has so far shown to be reliable for us when simply choosing a set of stocks to pick, but our primary goal with forecasting is to eventually forecast (or score) anything.

2.1 Measurement Theory

First we define a few mathematical terms to get our bearings. To the mathematical layperson, it is clear that physical objects have "measurable" dimensions, such as length and volume. Less obvious though, is the idea of assigning such a measure to a Probability Space.

In probability theory, we denote Ω to represent the entire sample space of possible outcomes. A Probability measure is a function that assigns a real number (i.e a probability) to some subset of the total sample space.

$$\mathcal{P}(\Omega) -> \mathbb{R} \tag{5}$$

When first being presented, there are commonly 5 basic properties of probability measures that one would like to see satisfied:

1. Existence (a measure $u(A)$ can be assigned to each set $A \subset \Omega$)
2. Additivity (when 2 disjoint subsets are combined, their measures can be added)
3. Shift invariance ($u(A) = u(A + x)$ i.e the mathematical equivalent that the diameter of a cylinder does not change when you move it)
4. Measureof entire event space equals 1

5. Non-negativity

However, one can construct examples of measures that cannot satisfy all 5 at once. The easiest way around this is to eliminate the first: Existence. This then leads to the concept of a σ - algebra. We solve the problem by simply defining a set of sets that *can be measured*

The measure of the entire σ-algebra must be 1, the empty set must be included (and assigned the number 0).

One obtains a σ - algebra by first determining some sets that should be measurable (for your current domain). And then identifying the rest of the σ-algebra via obeying all of the following properteis:

1. The empty set of the algebra
2. With any set B its complement $B^c = \Omega - B \in \mathcal{F}$
3. Along with B_1, B_2.. the Union over all B_n is included

2.2 Scoring

Tilmann Gneiting and Adrian E. Raftery define Proper Scoring elegantly in their paper *Strictly Proper Scoring Rules, Prediction, and Estimation*. They consider probabilistic forecasts on a general sample space Ω. Let $\mathcal{A}$ be a σ-algebra of subsets of Ω, and let $\mathcal{P}$ be a convex class of probability measures on $(\Omega, \mathcal{A})$. A function defined on Ω and taking values in the extended real line, $\overline{\mathbb{R}} = [-\infty, \infty]$, is $\mathcal{P}$-quasi-integrable if it is measurable with respect to $\mathcal{A}$ and is quasi-integrable with respect to all $P \in \mathcal{P}$. A probabilistic forecast is any probability measure $P \in \mathcal{P}$. A scoring rule is any extended real-valued function $S : \mathcal{P} \times \Omega \to \overline{\mathbb{R}}$ such that $S(P, \cdot)$ is $\mathcal{P}$-quasi-integrable for all $P \in \mathcal{P}$. Thus if the forecast is P and ω materializes, the forecaster's reward is $S(P, \omega)$.

While the jargon here may at first feel intimidating, the intuitions behind the concepts are simple. For any event space Ω, a forecaster can output it's own probability measure $P \subset \mathcal{P}$, where they've restricted their probability measures to a convex class. Under a true probability distribution Q, and the materialization of a particular event w, a Score is simply a function that maps that those 3 things into a real number. We require it to be integratable because we often use the *expected* score for our comparisons and this means we must integrate over $dQ(w)$

2.3 Proper Scoring

A proper scoring rule then is defined as the following, for all $P, Q \in \mathcal{P}$

$$S(Q, Q) \geq S(P, Q) \tag{6}$$

A *strictly* proper scoring rule assigns equality only when P=Q. A forecaster rated on such scoring rules is encouraged to produce measures that are closer with reality.

In an LLM forecasting situation where we are dealing with arbitrary forecasts over ill-defined event spaces, defining a proper scoring rule becomes essential. How do we do that?

Theorem 1. *A regular scoring rule $S : \mathcal{P} \times \Omega \to \overline{\mathbb{R}}$ is proper relative to the class $\mathcal{P}$ if and only if there exists a convex, real-valued function G on $\mathcal{P}$ such that*

$$S(P, \omega) = G(P) - \int G^*(P, \omega)\, dP(\omega) + G^*(P, \omega)$$

In plain words, all this is saying that we have some function G that converts probability measures to a real number. Any subtangent to this function at P is a straight line at that point. If we take the above theorem and then average over some truth Q, it tells us that a proper scoring rule is simply one where we have some function G that is convex and where G(P) = S(P, P).